

Motor Control Board

Variant: DRAFT

2026-09-05

Rev + (Unreleased)

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TOP VIEW

kibot_Mmage_png_3d_viewer_angled_top

BOTTOM VIEW

kibot_Mmage_png_3d_viewer_angled_bottom

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for
informational design
notes.

DESIGN NOTE:
Example text for
debug notes.

DESIGN NOTE:
Example text for
cautionary design
notes.

DESIGN NOTE:
Example text for
critical design
notes.

LAYOUT NOTE:
Example text for
critical layout
guidelines.

NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Block Diagram

File: Block Diagram.kicad_sch

Revision History

File: Revision History.kicad_sch

Project Architecture

File: Project Architecture.kicad_sch

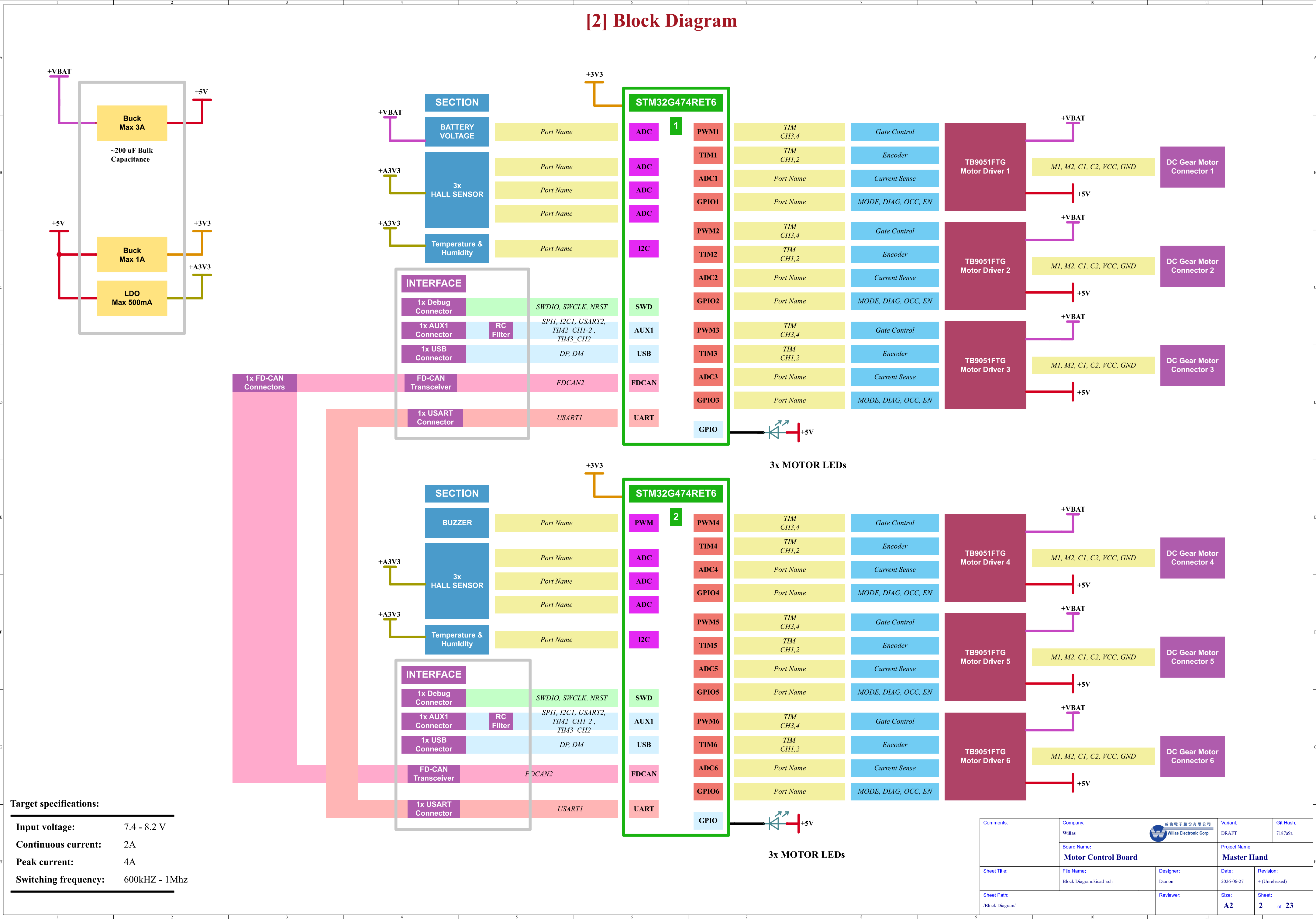
Power - Sequencing

File: Power - Sequencing.kicad_sch

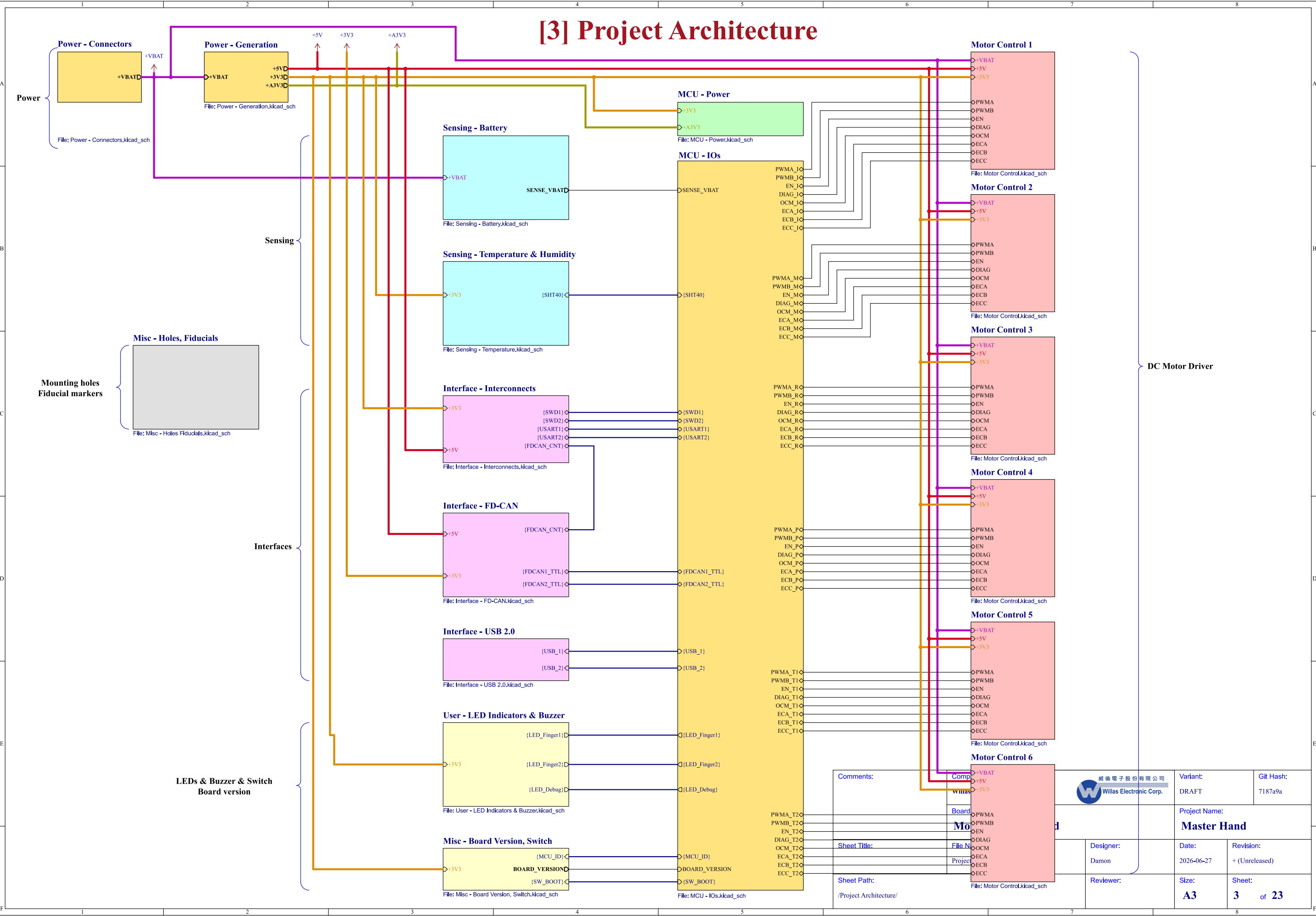
Date: 05-Sep-2026

Comments:	Company: Willas W Willas Electronic Corp.		Variant: DRAFT		Git Hash: 7187a9a	
	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: Master_Hand_Kicad.kicad_sch		Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /			Reviewer:	Size: A3	Sheet: 1 of 23	

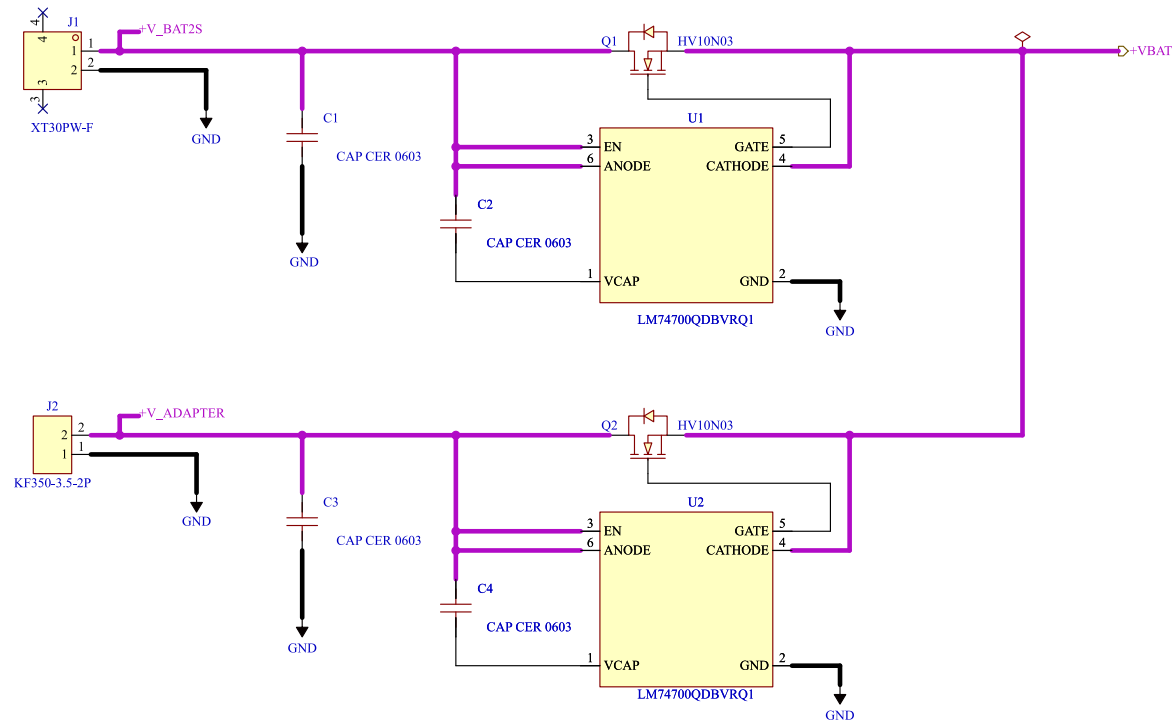
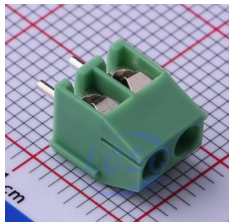
[2] Block Diagram



[3] Project Architecture



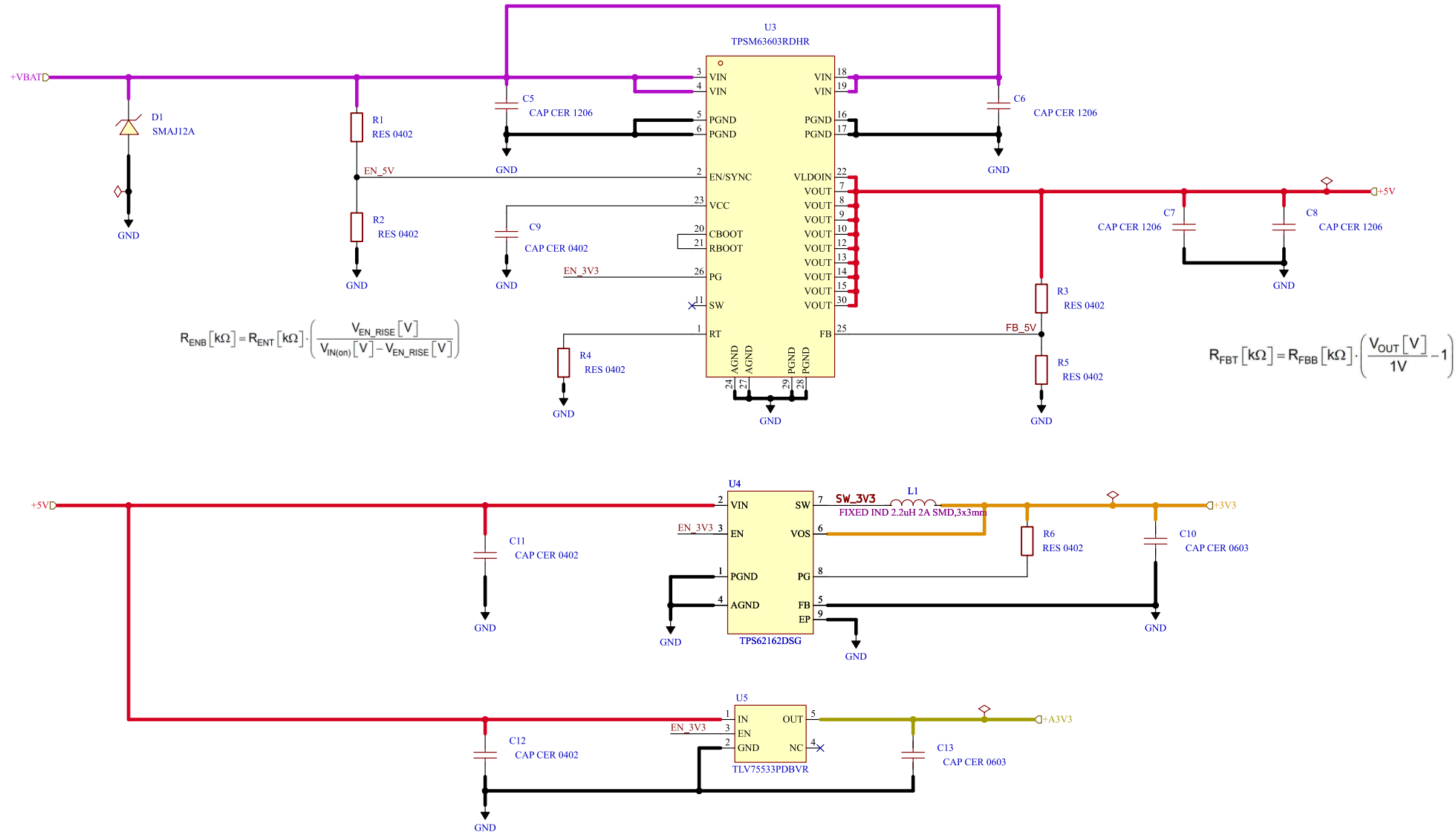
[4] Power - Connectors



Power Connectors - Path Control

Comments:	Company: Willas 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT		Git Hash: 7187a9a	
	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: Power - Connectors.kicad_sch		Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Power - Connectors/			Reviewer:	Size: A4	Sheet: 4 of 23	

[5] Power - Generation



DESIGN NOTE:
PI filter with inductor would give better EMC performance but has higher bulk capacitance requirements. Ferrite is a good compromise.

LAYOUT NOTE:
Route feedback traces away from switching node.

LAYOUT NOTE:
Minimize switching node loop areas.

Voltage Regulators

DC BULK

DC-Link Bulk Capacitors. 330uF + 10 x 10 uF = 430uF (35V)

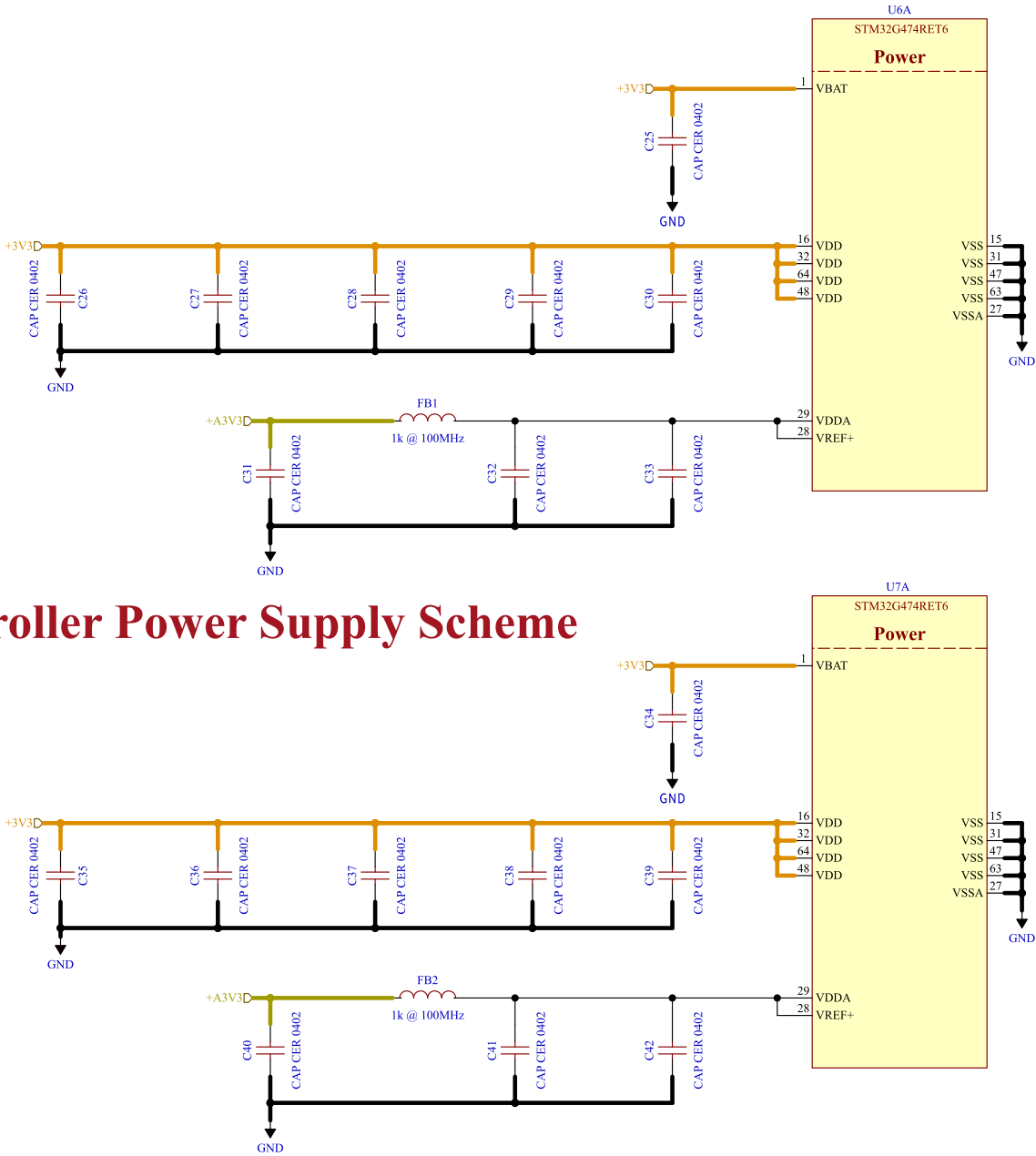
Test Points



Comments:	Company: Willas		 Willas Electronic Corp.	Variant: DRAFT	Git Hash: 7187a9a
	Board Name: Motor Control Board			Project Name: Master Hand	
Sheet Title:	File Name: Power - Generation.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Power - Generation/	Reviewer:		Size: A3	Sheet: 5 of 23	

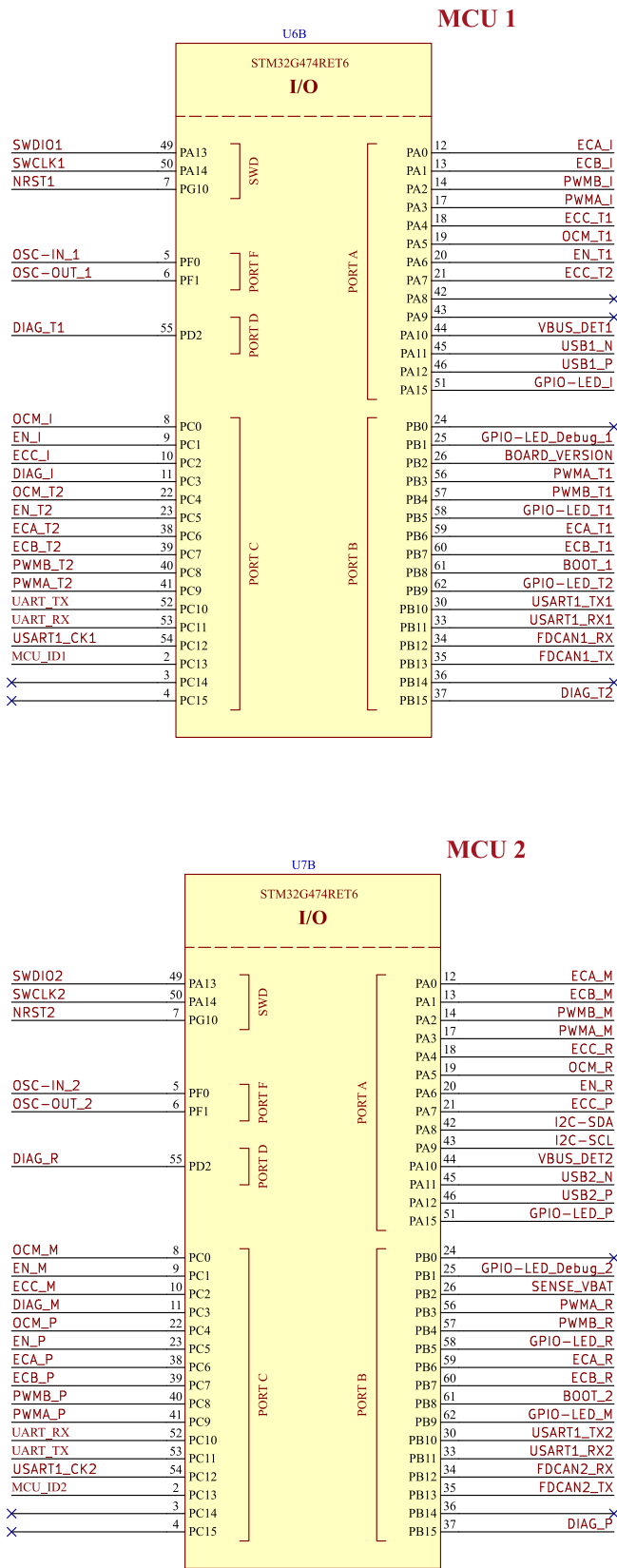
Microcontroller Power Supply Scheme

LAYOUT NOTE:
Place each capacitor close
to the relevant pin.
Minimize loop inductance.



Comments:	Company: Willas		W 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: 7187a9a
	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: MCU - Power.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/MCU - Power/		Reviewer:	Size: A3	Sheet: 6 of 23		

[7] MCU - IOs



DESIGN NOTE:

We use the LQFP64 version to have extra pins for controlling an external fan.

DESIGN NOTE:

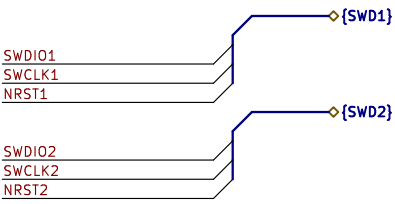
Do not forget to flash option bytes when flashing for the first time.

LAYOUT NOTE:

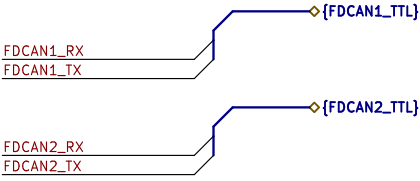
Footprint deviates from datasheet and IPC7351 to allow for soldermask sliver between pads according to PCBWay design rules.

SENSE_VBAT → SENSE_VBAT

Battery Voltage Sense



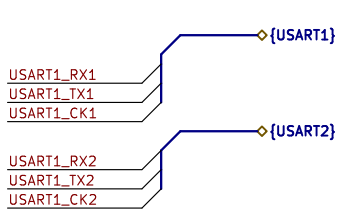
SWD (Debug) Port



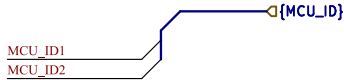
5Mbps FD-CAN Bus

BOARD_VERSION → BOARD_VERSION

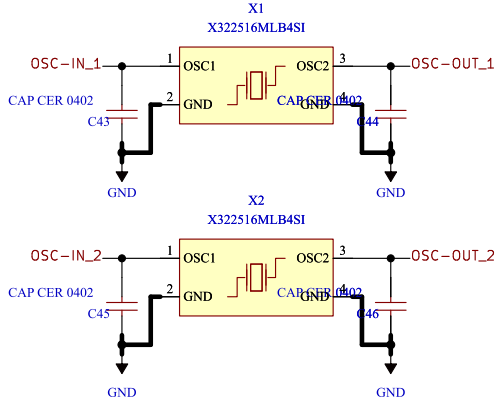
Board Version Selection



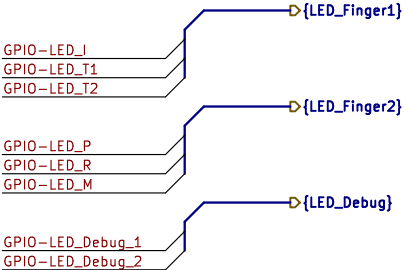
MCB to MB - USART



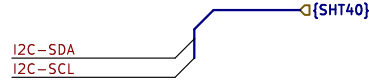
MCU ID



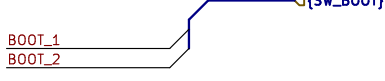
Oscillator HSE



LEDs



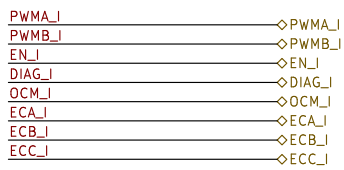
Temperature and Humidity



SWITCH



USB PORT



Motor Driver

Battery Voltage Sense

SENSE_VBAT → TP8

Board Version Selection

BOARD_VERSION → TP12

SWD (Debug) Port

NRST2 → TP16
SWCLK2 → TP18
SWDIO2 → TP20

5Mbps FD-CAN Bus

FDCAN1_TX → TP28
FDCAN1_RX → TP30
FDCAN2_TX → TP33
FDCAN2_RX → TP35

LEDs

GPIO-LED_I → TP37
GPIO-LED_T1 → TP39
GPIO-LED_T2 → TP41
GPIO-LED_R → TP43
GPIO-LED_M → TP45
GPIO-LED_Debug_1 → TP49
GPIO-LED_Debug_2 → TP51

Temperature and Humidity

I2C-SDA → TP53
I2C-SCL → TP55

USB PORT

USB1_P → TP63
USB1_N → TP65
VBUS_DET1 → TP66

MCB to MB - I2C

I2C_SCL → TP76
I2C_SDA → TP78

Buzzer

MCU ID

MCU_ID1 → TP84
MCU_ID2 → TP86

Test Points

MD Index

PWMA_I → TP9
PWMB_I → TP10
EN_I → TP11
DIAG_I → TP13
OCM_I → TP14
ECA_I → TP15
ECB_I → TP17
ECC_I → TP19

MD Middle

PWMA_M → TP22
PWMB_M → TP24
EN_M → TP26
DIAG_M → TP27
OCM_M → TP29
ECA_M → TP31
ECB_M → TP32
ECC_M → TP34

MD Ring

PWMA_R → TP36
PWMB_R → TP38
EN_R → TP40
DIAG_R → TP42
OCM_R → TP44
ECA_R → TP46
ECB_R → TP48
ECC_R → TP50

MD Pinky

PWMA_P → TP52
PWMB_P → TP54
EN_P → TP56
DIAG_P → TP58
OCM_P → TP60
ECA_P → TP61
ECB_P → TP62
ECC_P → TP64

MD Thumb 1

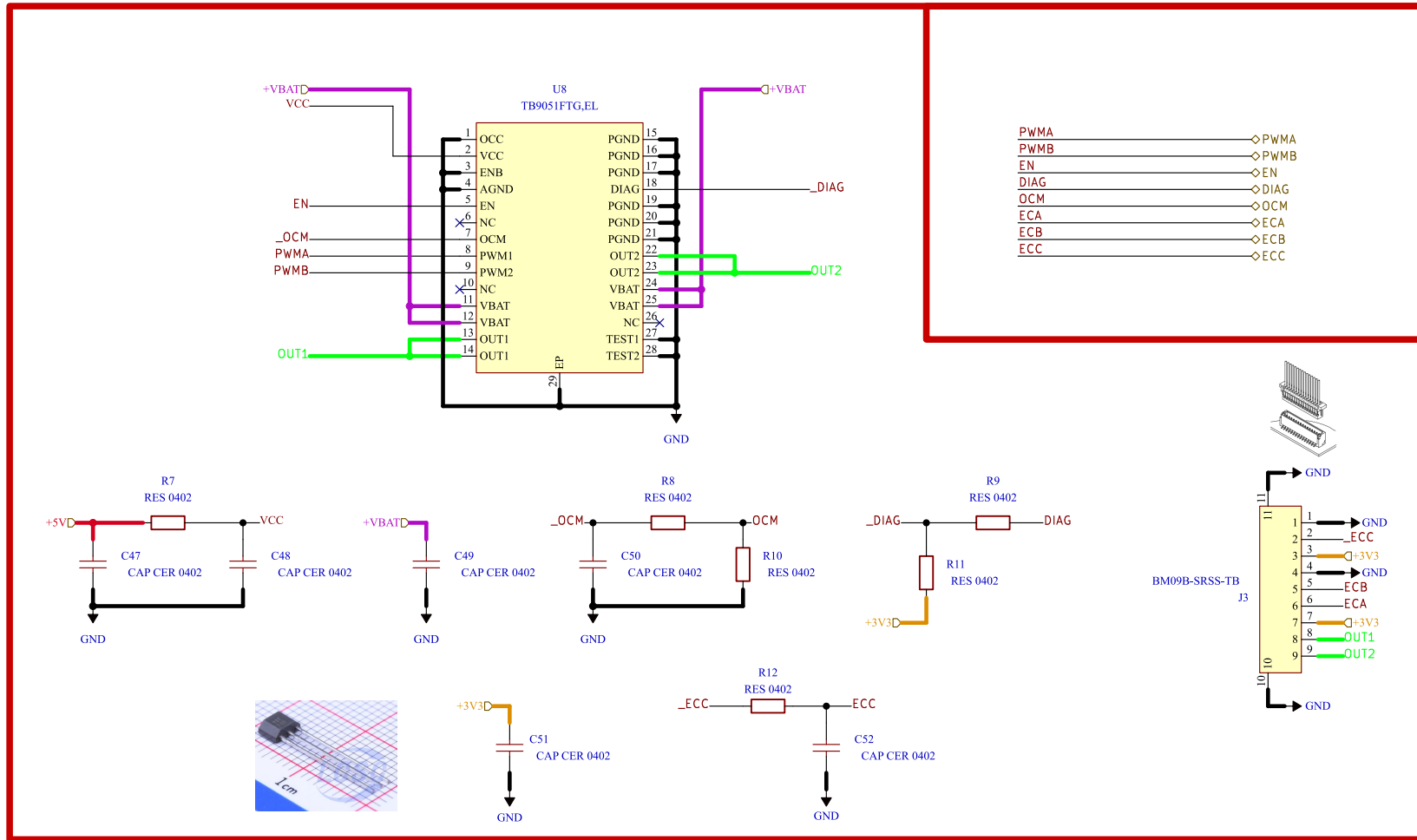
PWMA_T1 → TP67
PWMB_T1 → TP69
EN_T1 → TP71
DIAG_T1 → TP73
OCM_T1 → TP74
ECA_T1 → TP75
ECB_T1 → TP77
ECC_T1 → TP79

MD Thumb 2

PWMA_T2 → TP80
PWMB_T2 → TP81
EN_T2 → TP82
DIAG_T2 → TP83
OCM_T2 → TP85
ECA_T2 → TP87
ECB_T2 → TP88
ECC_T2 → TP89

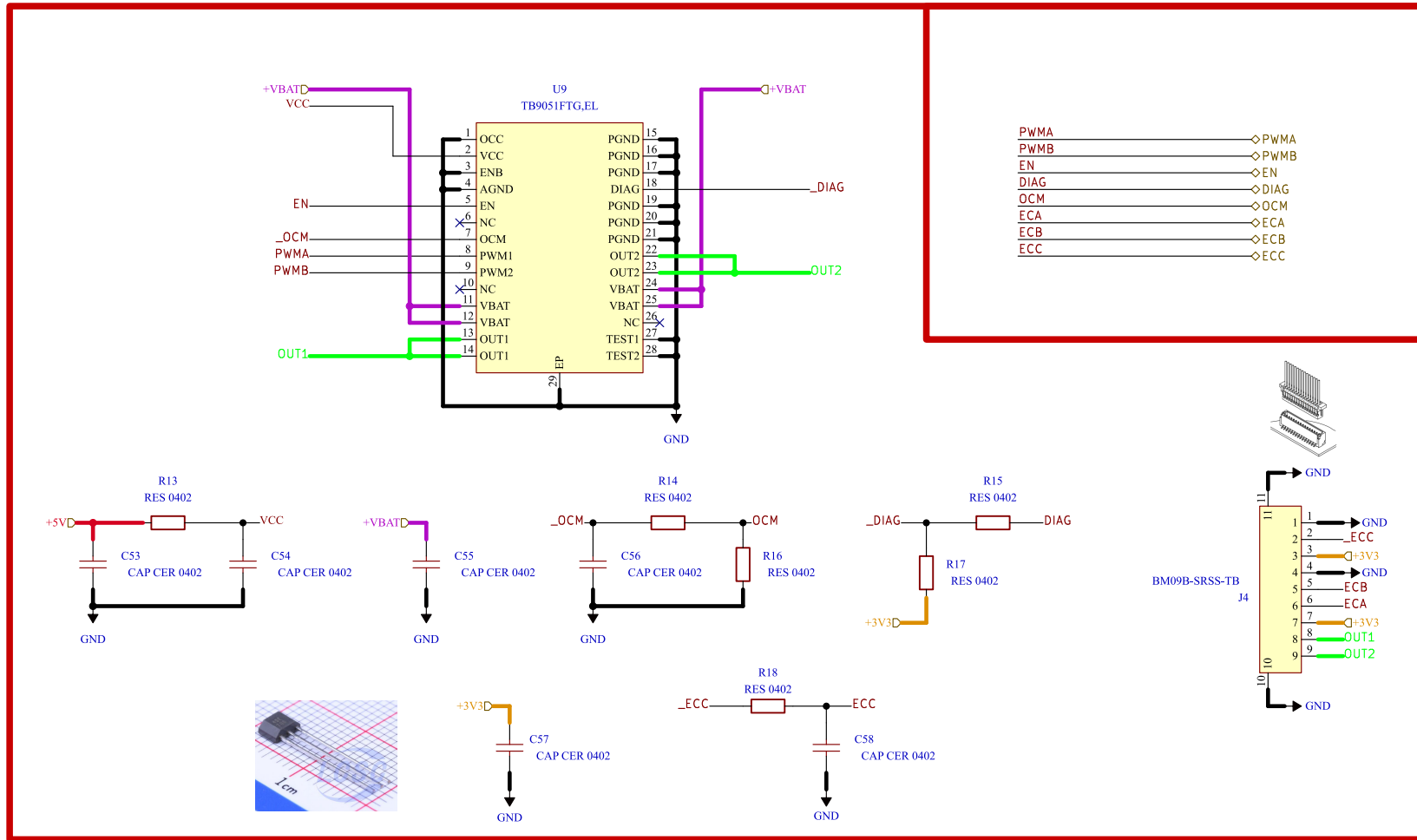
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	Board Name: Motor Control Board				Project Name: Master Hand	
Sheet Title:	File Name: MCU - IOs.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/MCU - IOs/			Reviewer:	Size: A3	Sheet: 7 of 23	

[8] Motor Control 1



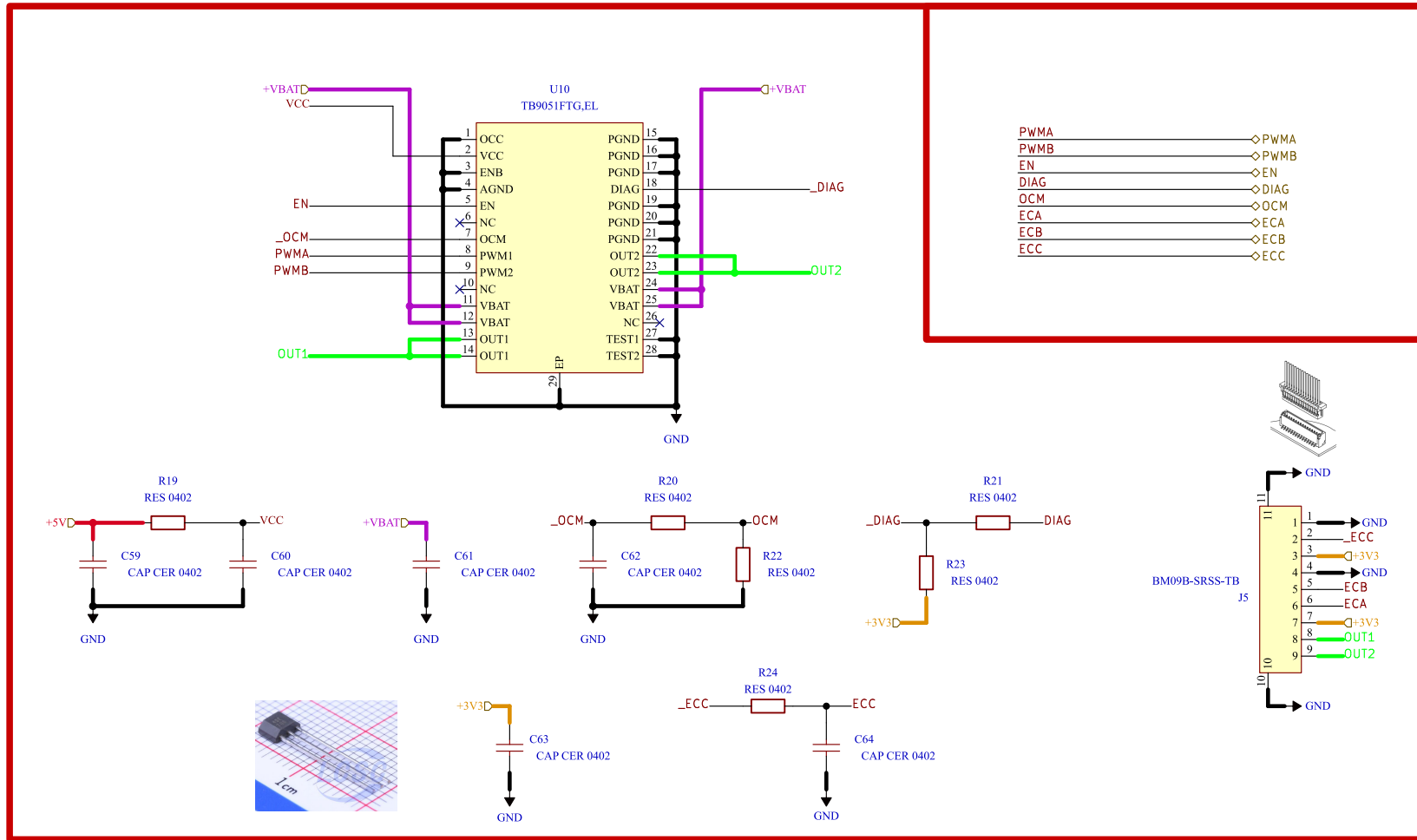
Comments:	Company: 威倫電子股份有限公司 Willas		Variant: DRAFT	Git Hash: 7187a9a
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control 1/	Reviewer:	Size: A4	Sheet: 8 of 23	

[9] Motor Control 2



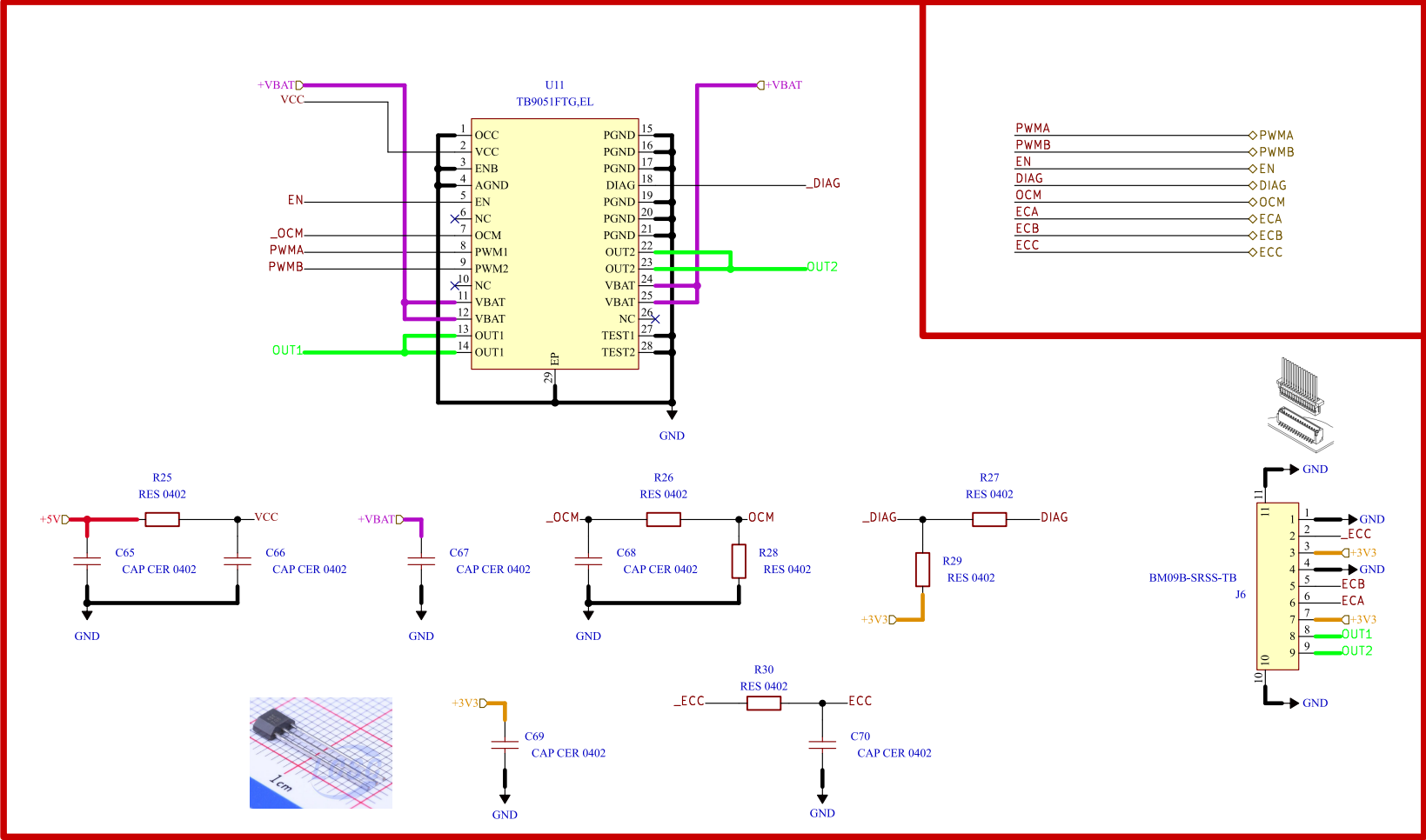
Comments:	Company: 威倫電子股份有限公司 Willas		Variant: DRAFT	Git Hash: 7187a9a
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control 2/	Reviewer:	Size: A4	Sheet: 9	of 23

[10] Motor Control 3



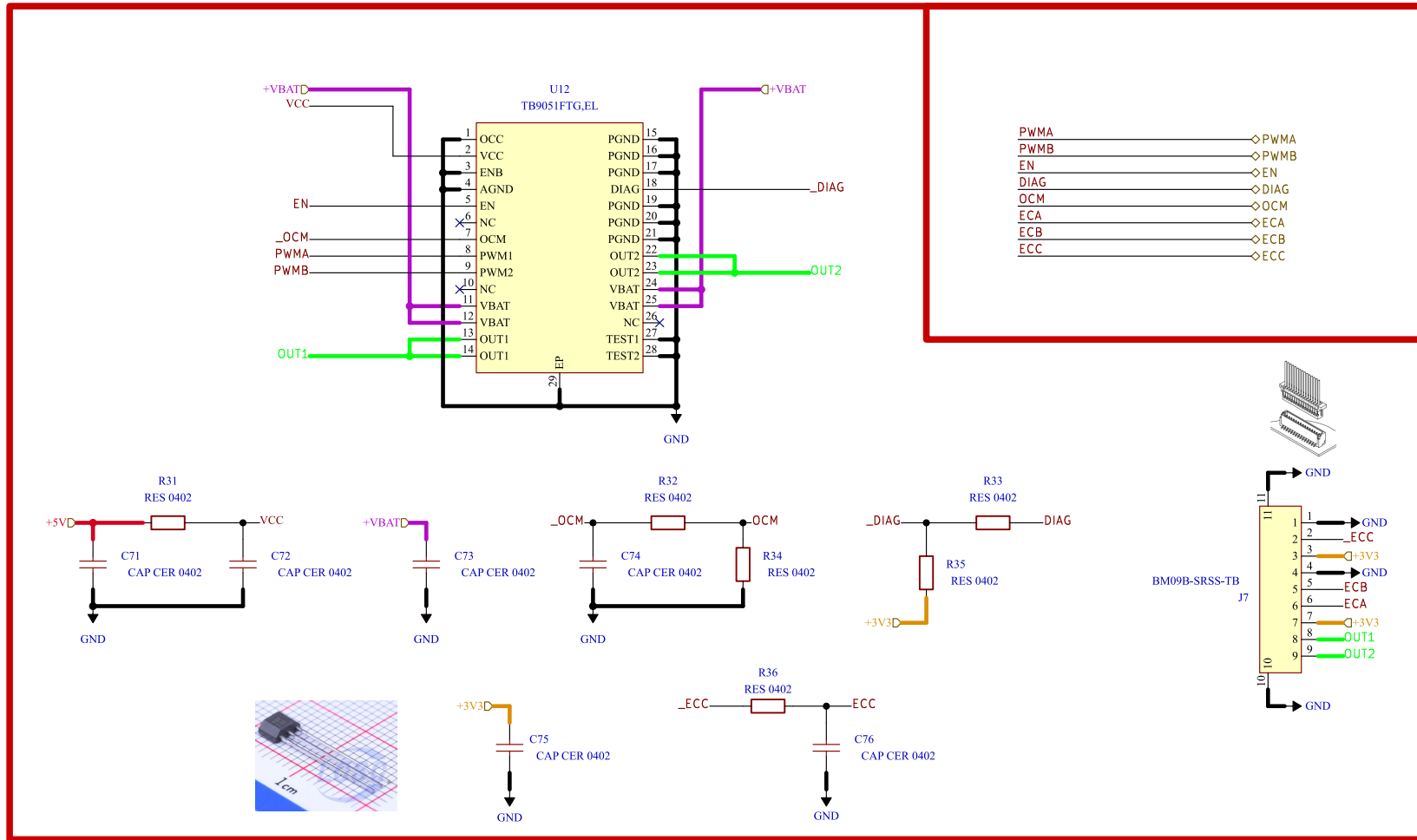
Comments:	Company: Willas 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: 7187a9a
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control 3/		Reviewer:	Size: A4	Sheet: 10 of 23

[11] Motor Control 4



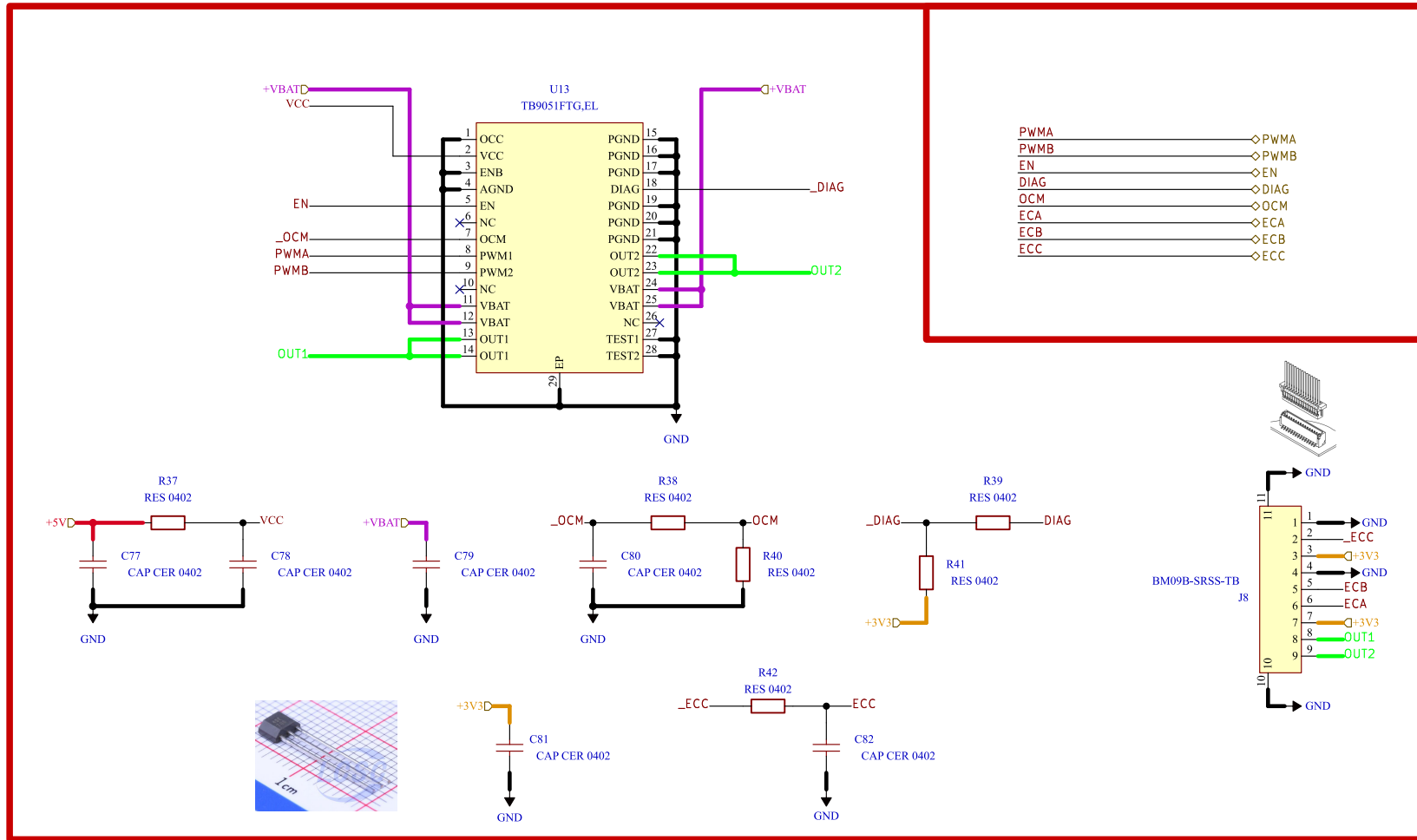
Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT		Git Hash: 7187a9a
	Board Name: Motor Control Board		Project Name: Master Hand		
Sheet Title:	File Name: Motor Control.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Motor Control 4/		Reviewer:	Size: A4	Sheet: 11 of 23	

[12] Motor Control 5



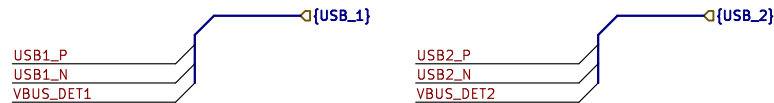
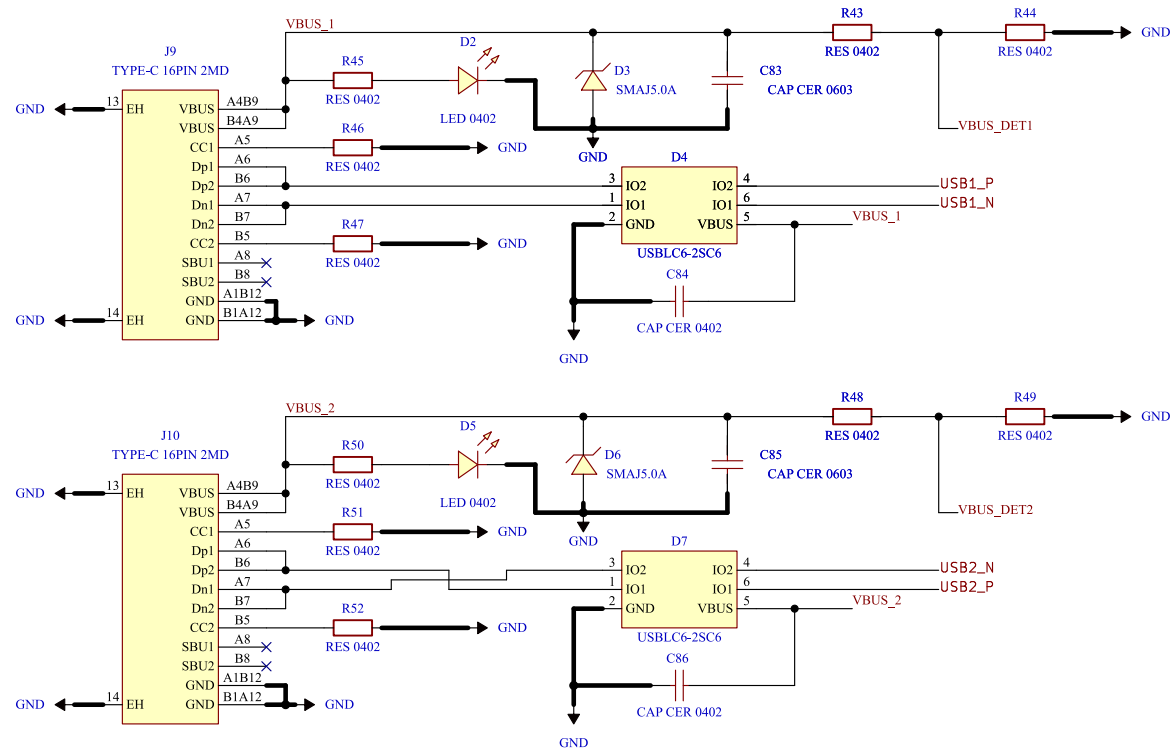
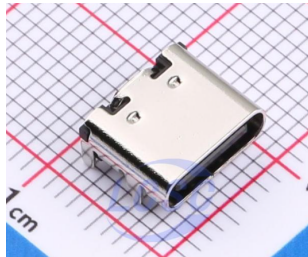
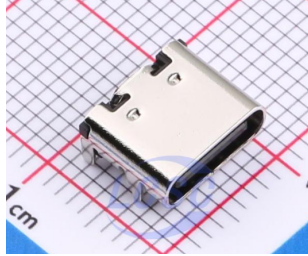
Comments:	Company: Willas 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: 7187a9a
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control 5/		Reviewer:	Size: A4	Sheet: 12 of 23

[13] Motor Control 6



Comments:	Company: Willas 		Variant: DRAFT	Git Hash: 7187a9a
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control 6/	Reviewer:	Size: A4	Sheet: 13 of 23	

[14] Interface - USB 2.0



DESIGN NOTE:

- [1] VBUS NOT connected to +5V/+3V3. Board is self-powered(LiPo). Sense + LED only. No backfeed to host.
- [2] LED direct from VBUS @ 1mA. USB2.0 suspend limit 2.5mA. LED 1mA + divider 33uA = 1.03mA (41%). Do NOT raise to 5mA.
- [3] VBUS_DET required - G474 has no dedicated VBUS pin. Self-powered device must not pull up D+ when VBUS=0. FW controls DPPU bit in USB_BCDR. 68k/100k: VBUS 4.40V -> 2.60V > VIH 2.31V. Debounce 100ms.
- [4] D+/D- direct to PA12/PA11. No 22R series. No external 1k5 pull-up (internal RPU1 900-1500R).
- [5] USB clock: PLL-Q 48MHz from HSE 8MHz. ST DFU bootloader uses HS148+CRS - flashable without crystal.
- [6] Both USB ports fully isolated - no shared nets.
- [7] GROUND LOOP WARNING: USB from PC ties host GND to power GND (4.6A peak @20kHz). Use isolator (ADuM3160) or isolated USB-UART when debugging with motors running.

Comments:

Company:

Willas



Variant:

DRAFT

Git Hash:

7187a9a

Board Name:

Motor Control Board

Project Name:

Master Hand

Sheet Title:

File Name:

Interface - USB 2.0.kicad_sch

Designer:

Damon

Date:

2026-06-27

Revision:

+(Unreleased)

Sheet Path:

/Project Architecture/Interface - USB 2.0/

Reviewer:

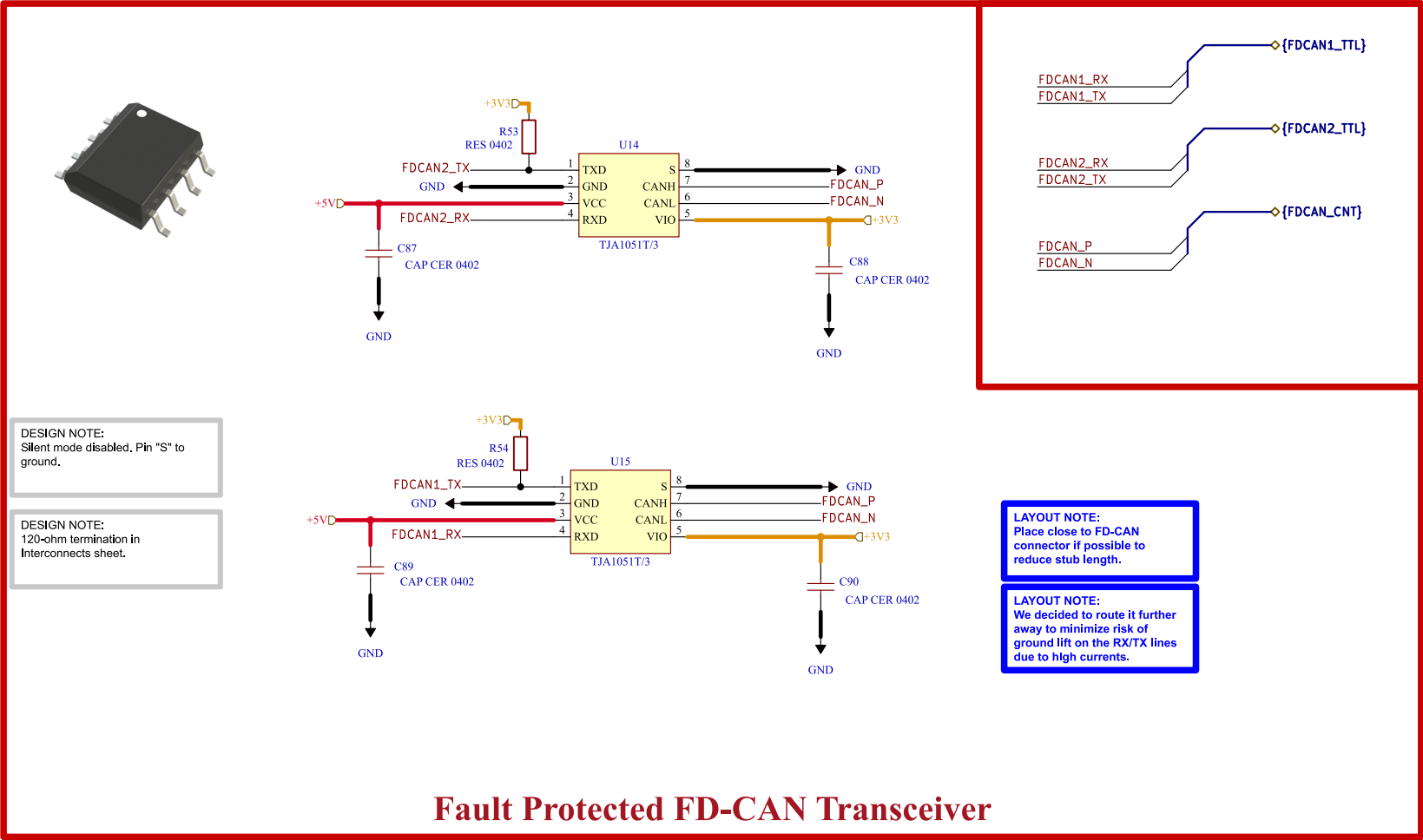
Size:

A4

Sheet:

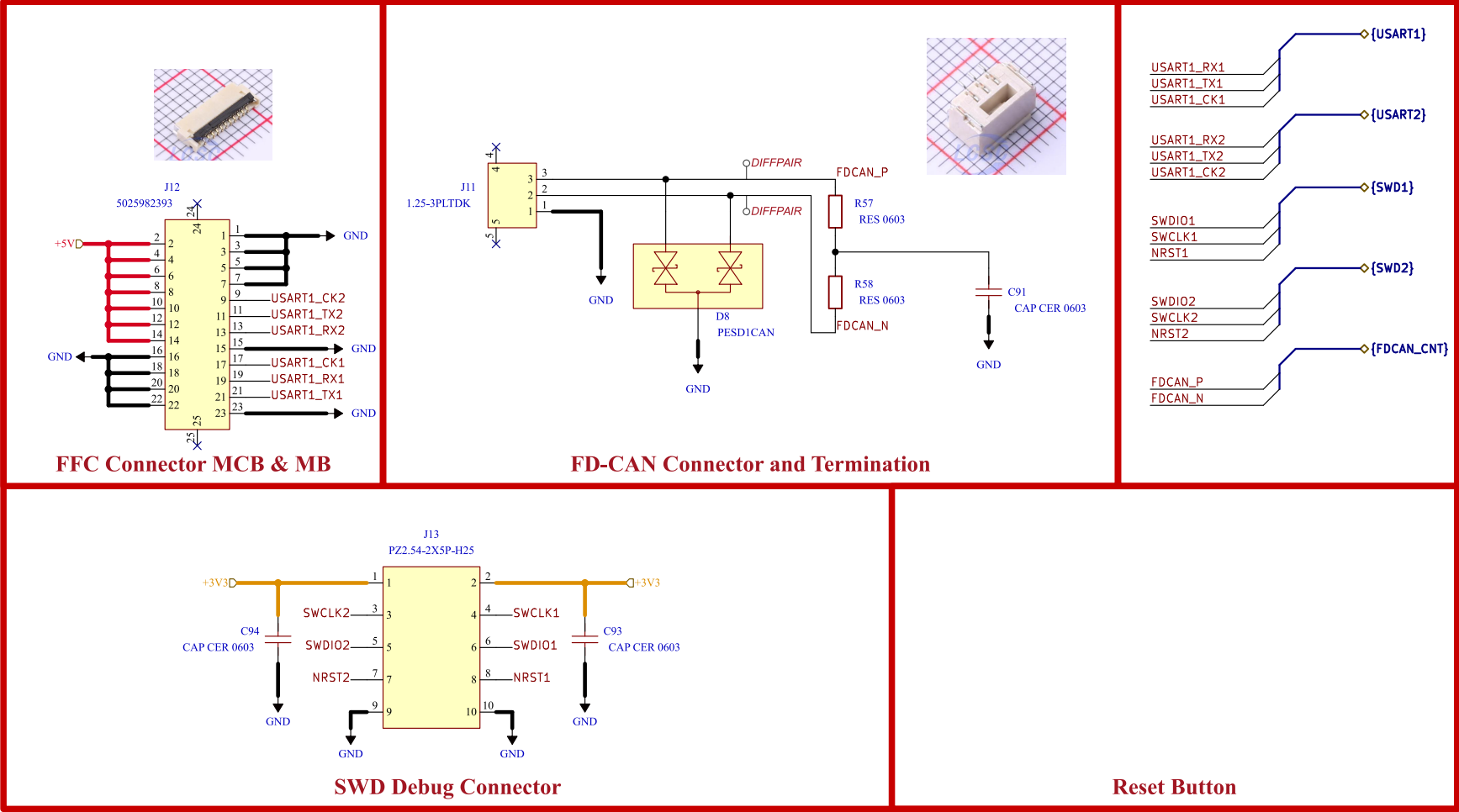
14 of 23

[15] Interface - FD-CAN



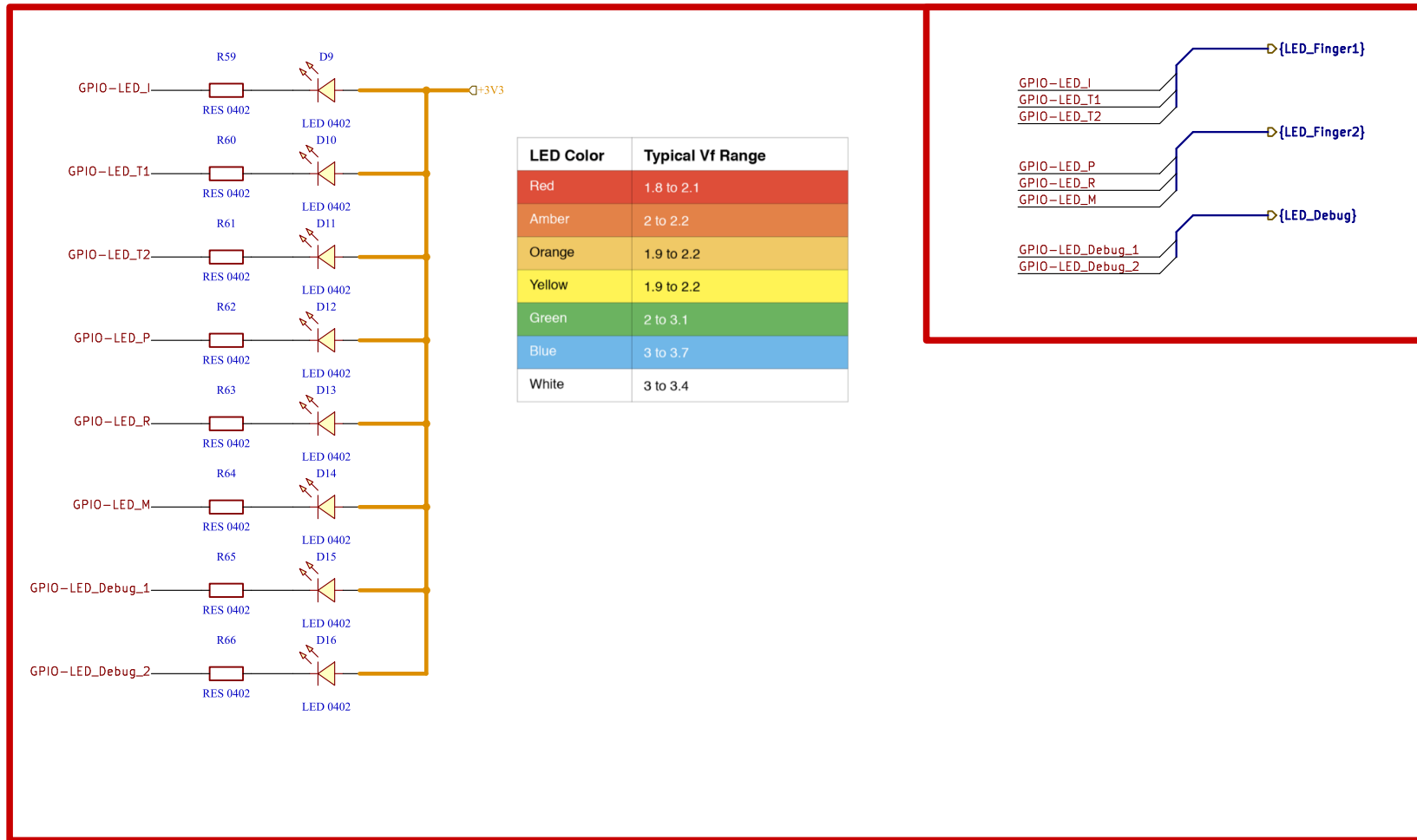
Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT		Git Hash: 7187a9a
	Board Name: Motor Control Board			Project Name: Master Hand	
Sheet Title:	File Name: Interface - FD-CAN.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Interface - FD-CAN/		Reviewer:	Size: A4	Sheet: 15 of 23	

[16] Interface - Interconnects



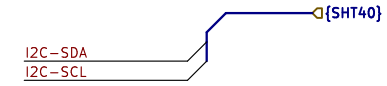
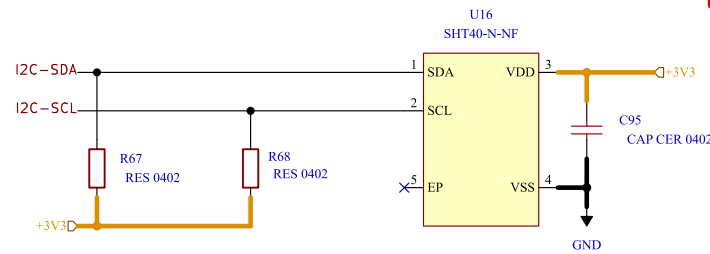
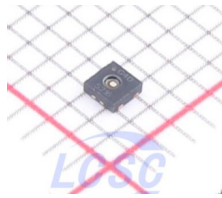
Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT		Git Hash: 7187a9a	
	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: Interface - Interconnects.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/Interface - Interconnects/		Reviewer:	Size: A4	Sheet: 16 of 23		

[17] User - LED Indicators & Buzzer



Comments:	Company:  威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: 7187a9a
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: User - LED Indicators & Buzzer.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/User - LED Indicators & Buzzer/		Reviewer:	Size: A4	Sheet: 17 of 23

[18] Sensing - Temperature & Humidity



Comments:	Company: Willas		Variant: DRAFT	Git Hash: 7187a9a
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Sensing - Temperature.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensing - Temperature & Humidity/		Reviewer:	Size: A4	Sheet: 18 of 23

[19] Sensing - Battery

DESIGN NOTE:
Battery voltage sensing divider for 2S LiPo battery.

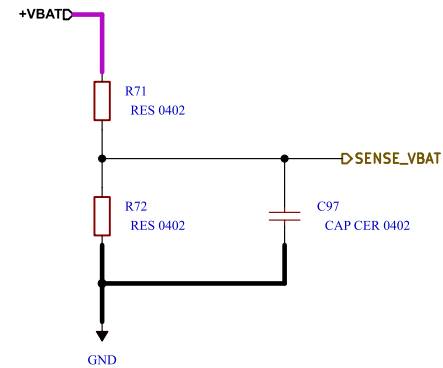
R1 = 100kΩ (Top)
R2 = 47kΩ (Bottom)

Divider Ratio:
 $V_{ADC} = V_{BAT} \times (47 / (100 + 47))$
 $V_{ADC} = V_{BAT} \times 0.3197$

2S LiPo:
Max battery voltage (8.4V): 2.69 V
Nominal voltage (7.4V): 2.37 V
Low battery voltage (6.4V): 2.05 V
Cutoff voltage (6.0V): 1.92 V

Maximum divider current:
 $I = 8.4V / 147k\Omega = 57 \mu A$

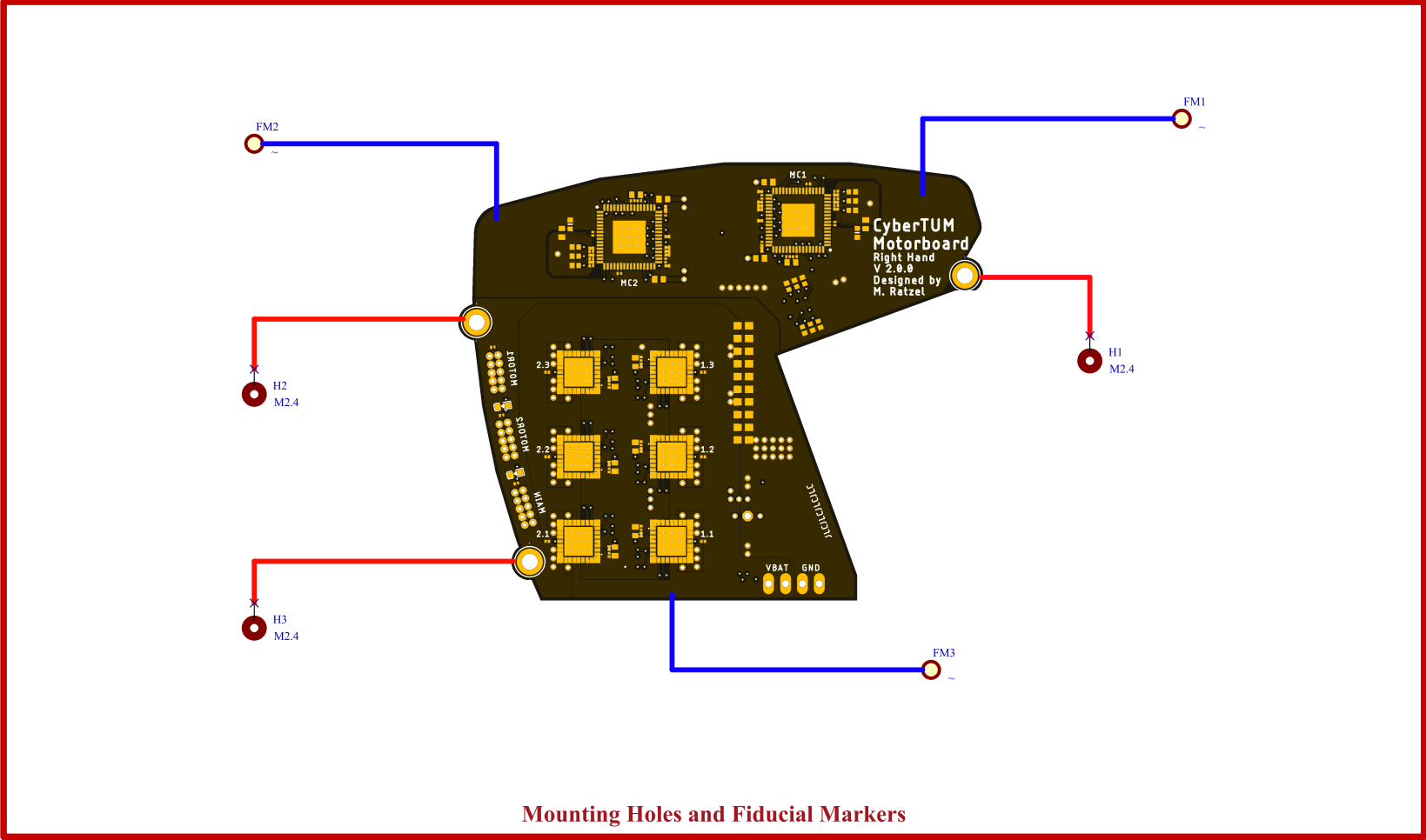
Notes:
- Designed for 3.3V ADC input.
- Add 100nF capacitor at ADC input for noise filtering.
- Use 1% resistors or better.
- Use low temperature coefficient resistors ($\leq 100 \text{ ppm}/^\circ\text{C}$) for improved measurement accuracy.



Battery Voltage Sensing

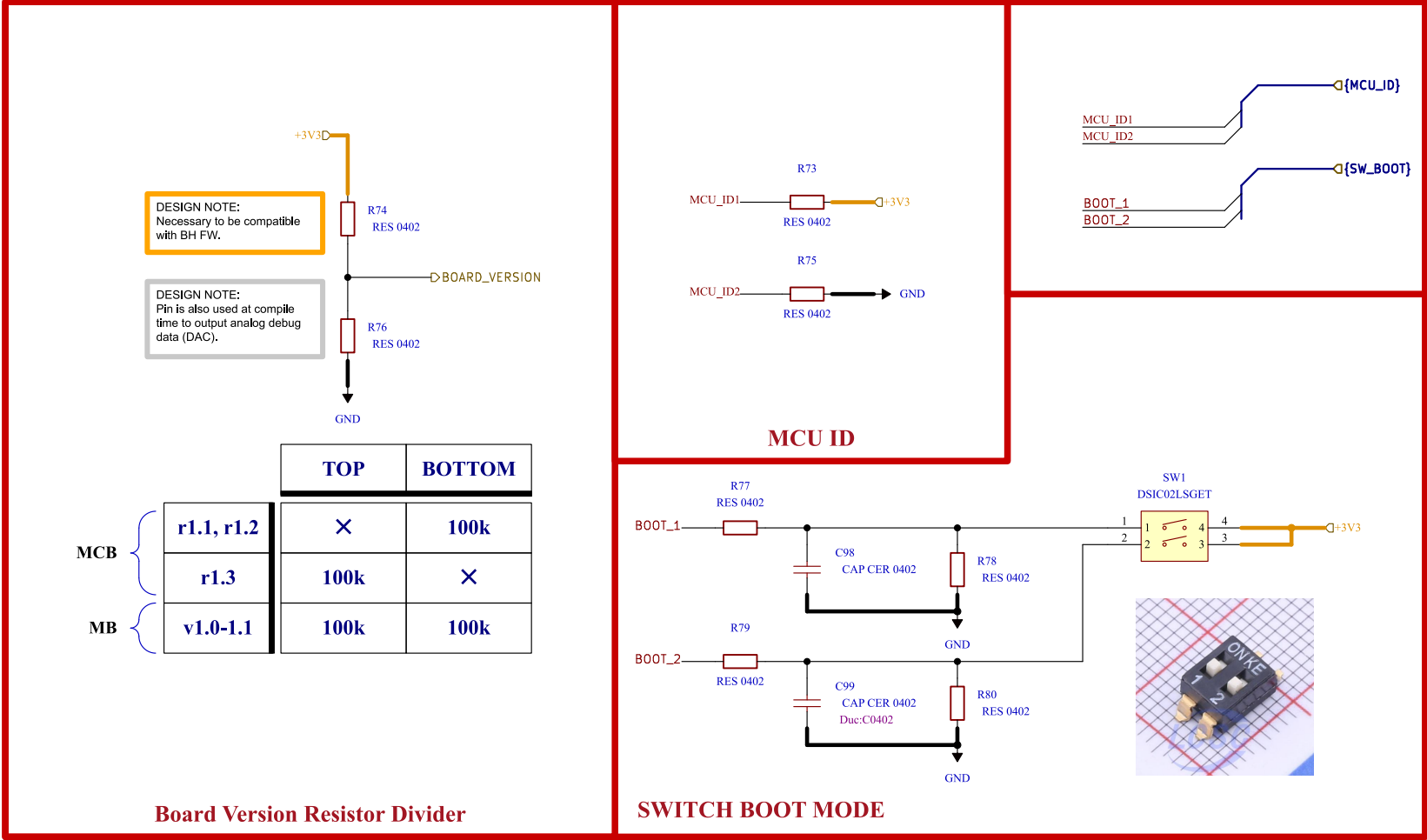
Comments:	Company: Willas 		Variant: DRAFT	Git Hash: 7187a9a
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Sensing - Battery.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensing - Battery/		Reviewer:	Size: A4	Sheet: 19 of 23

[20] Misc - Holes, Fiducials



Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT	Git Hash: 7187a9a
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:	Size: A4	Sheet: 20 of 23

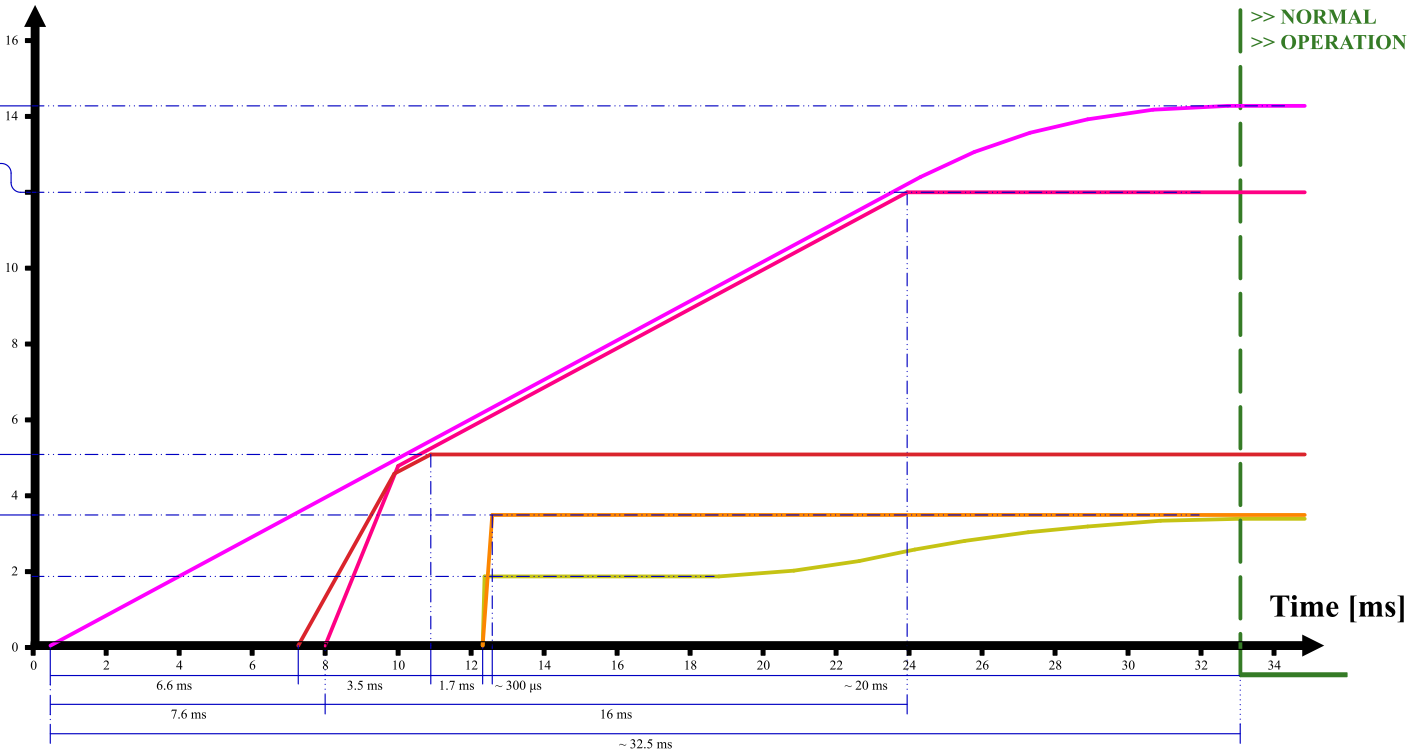
[21] Misc - Board Version, Switch



Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT		Git Hash: 7187a9a
	Board Name: Motor Control Board			Project Name: Master Hand	
Sheet Title:	File Name: Misc - Board Version, Switch.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Misc - Board Version, Switch/		Reviewer:	Size: A4	Sheet: 21 of 23	

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%

Voltage [V]



Comments:	Company: Willas 		Variant: DRAFT	Git Hash: 7187a9a
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Power - Sequencing.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 22 of 23

Comments:	Company: Willas  威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT		Git Hash: 7187a9a
	Board Name: Motor Control Board			Project Name: Master Hand	
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Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 23 of 23	